

Effect of Climate Scenarios on Pear Production under Controlled Ecotron Units

Project: Multivariate and Hierarchical Data (3565)

Project Discovering Associations

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Abstract

This project investigates the effect of four scenarios projected for 2070 on pear yield in a controlled experiment conducted in 24 ecotrons from Belgium and France. Each ecotron unit was observed with two varieties (Conference and Doyenne du Comice), and the climate scenarios were: no intervention (scenario 1), active CO₂ removal (scenario 2), sustainable production, and sustainable transportation. Pear yield was modelled with a Poisson Generalised Linear Mixed Model which included climate scenario, varieties, and location as fixed effects and random effect for the ecotron. Climate scenario had a highly significant effect on yield ($\chi^2(3) = 47.02$, $p < 0.001$). The CO₂-removal scenario produced the highest mean yield (26.0 pears/tree), representing a 71% increase over the no-intervention baseline (rate ratio = 1.71, $p < 0.001$), while the sustainable energy scenario yielded 32% more than the baseline (rate ratio = 1.32, $p < 0.001$). The transportation scenario was indistinguishable from the control ($p = 0.985$). These results, consistent with a parallel analysis of pear quality, identify CO₂ removal as the most beneficial climate intervention for pear production and growth. An exploratory multivariate analysis giving insights on the role of soil as a mediator between climate and production is also presented.

Key Words: ecotron units; Poisson GLMM; climate change; pear production; soil mediator.

1 Introduction

Climate change is expected to substantially affect fruit production. Pear cultivation in Belgium in particular is expected to face increasing challenges in maintaining stable yield, fruit quality, and sustainable production. To investigate potential future outcomes, this study evaluates the effect of several climate intervention scenarios projected for the year 2070 on pear production. Four climate scenarios were considered, representing different intervention strategies including no major intervention (Scenario 1), active CO₂ removal (Scenario 2), sustainable energy transition (Scenario 3), and transportation-focused interventions (Scenario 4). Two pear varieties commonly cultivated in Belgium, Conference and Doyenne du Comice, were included in the experiment. Pear trees were grown under controlled Ecotron conditions to simulate future climates while minimizing uncontrolled environmental variation.

The primary objective of this study was to assess which future climate intervention scenario is most favorable for pear production quality under projected climate conditions in 2070. A secondary objective is to investigate the effect of climate conditions on pear size trajectories over time. Because pear growth is inherently longitudinal and repeated measurements were collected from the same pears throughout the study period, the analysis needed to account for within-pear dependence and potential confounding introduced by the experimental design. For the longitudinal growth analysis, the primary research

question was whether pear growth trajectories differed across climate scenarios and pear varieties over time. The following hypothesis was considered: H_0 : Pear growth trajectories do not differ between climate scenarios. An additional hypothesis regarding varieties differences was also investigated: H_0 : Conference and Doyenne pears exhibit identical growth trajectories.

A complementary multivariate analysis is conducted to explore the relationships between climate scenarios, soil properties (at the end of the experiment), and pear production outcomes. Since soil conditions were homogeneous at the beginning of the experiment, the final soil variables are not interpreted as baseline covariates or confounding factors, but rather as potential components of the system response to the climate treatments (mediator).

2 Methods and Materials

2.1 Primary Objective: Pear Yield

2.1.1 Design

The primary objective is to identify the best scenario for the yield of pear compared to scenario 1. For this, in total, 24 ecotrons were used: 12 in Belgium and France, and the scenarios were randomized at the ecotron level, with three ecotrons per scenario per country. Within each ecotron, six pear trees were planted (three of each variety), resulting in 144 trees. The primary outcome of the main primary objective is the number of pears harvested per tree at the end of the experiment. The count result is analyzed separately for pear quality and pear size over time.

2.1.2 Statistical Model

Let Y_{ijkl} denote the pear count of the k -th tree ($k = 1, \dots, 6$) in ecotron j ($j = 1, \dots, 6$) assigned to climate scenario i ($i = 1, 2, 3, 4$) at location l ($l = \text{BE}, \text{FR}$). We model

$$Y_{ijkl} \mid b_{jl} \sim \text{Poisson}(\mu_{ijkl}), \quad \log(\mu_{ijkl}) = \eta_{ijkl}, \quad (1)$$

where the linear predictor is

$$\eta_{ijkl} = \beta_0 + \beta_i^{\text{clim}} + \beta^{\text{sp}} \cdot \mathbf{1}[\text{species}_{jk} = \text{Doyenne}] + \beta^{\text{loc}} \cdot \mathbf{1}[l = \text{FR}] + b_{jl}. \quad (2)$$

The parameters are:

- β_0 : overall intercept (log mean count for Scenario 1, *Conference*, Belgium);
- β_i^{clim} , $i = 2, 3, 4$: fixed contrasts for climate scenarios relative to Scenario 1 (reference), with $\beta_1^{\text{clim}} \equiv 0$;
- β^{sp} : fixed effect of varieties (*Doyenne* vs. *Conference*);

- β^{loc} : fixed effect of location (France vs. Belgium);
- b_{jl} : random effect for ecotron j at location l , with $b_{jl} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_b^2)$.

The random intercept for ecotron accounts for the clustering of six trees within the same experimental unit, which induces within-ecotron correlation. The theoretical framework for GLMMs of this form is developed in [1, Ch. 14, pp. 265–279]; foundations of Poisson regression are in [1, Ch. 3, pp. 27–33]. The model was fitted using the `lme4` package [2] using the `glmer()` method.

Exponentiated coefficients $\exp(\beta_i^{\text{clim}})$ are interpreted as rate ratios: the expected pear count under Scenario i relative to Scenario 1, after adjusting for varieties and location. A likelihood ratio test (LRT) comparing the full model to a model without climate fixed effects was used to assess the overall significance of the climate effect ($\alpha = 0.05$). Model adequacy was assessed by inspecting Pearson residuals and by computing the ratio of the Pearson chi-squared statistic to its degrees of freedom (a ratio near 1 indicates no residual over-dispersion). A Negative Binomial model was fitted as a robustness check.

2.1.3 Research Hypotheses

The research question is whether the choice of climate intervention affects pear yield. The primary statistical hypothesis is an omnibus test of no climate effect on the expected number of pears per tree:

$$H_0 : \beta_2^{\text{clim}} = \beta_3^{\text{clim}} = \beta_4^{\text{clim}} = 0, \quad (3)$$

$$H_1 : \beta_i^{\text{clim}} \neq 0 \text{ for at least one } i \in \{2, 3, 4\}, \quad (4)$$

where the parameters are defined in the statistical model below. Rejection of H_0 establishes that climate scenario has an overall effect on pear yield, but does not identify which scenarios differ. Conditional on rejecting H_0 , pairwise secondary hypotheses of the form

$$H_0^{(i,i')} : \beta_i^{\text{clim}} = \beta_{i'}^{\text{clim}}, \quad i \neq i', \quad i, i' \in \{1, 2, 3, 4\}, \quad (5)$$

are tested using estimated marginal means [3] with Tukey adjustment for multiplicity, to locate the source of the difference and to identify the scenario with the highest expected yield.

2.1.4 Sample Size Calculation

The minimum number of ecotrons needed to detect a biologically relevant difference in pear yield was determined via a simulation-based power analysis [4]. Parameters required for the simulation (baseline yield λ , varieties effect, ecotron-level variability σ_b) were estimated from pilot data comprising $n = 24$ trees from four Belgian ecotrons under Scenario 1. A

Poisson GLMM with varieties as a fixed effect and ecotron as a random intercept was fitted to the pilot data, yielding $\hat{\lambda}_0 = 3.75$ (log scale; ≈ 42 pears per *Conference* tree), a Doyenne effect of $\hat{\beta}^{\text{sp}} = -0.84$ ($\approx 57\%$ fewer pears), and $\hat{\sigma}_b = 0.035$.

For the full-study power simulation, a minimum detectable scenario effect of $\exp(\delta) = 1.10$ (10% increase relative to Scenario 1) was specified in agreement with the research team. Each simulated replicate drew ecotron random effects from $\mathcal{N}(0, \hat{\sigma}_b^2)$, generated tree-level counts from the Poisson model, fitted the GLMM, and recorded whether the omnibus LRT for the climate effect was significant at $\alpha = 0.05$. Power was estimated as the proportion of significant tests across 100 simulations per sample-size configuration.

The power curve showed that six ecotrons per scenario (24 ecotrons total, 144 trees) achieved power of approximately 0.91, comfortably exceeding the pre-specified target of 0.85. This configuration, combined with the use of two countries as an additional blocking factor, was adopted as the final study design.

2.2 Pear Quality Analysis

Pear quality was analyzed using a quality index ranging from 0 to 100, with higher values indicating better quality. The primary objective of this analysis was to examine whether pear quality differs across the four climate scenarios.

Pear variety and ecotron location were also included in the analysis, as differences in quality could partly be attributed to the variety or to the country where the ecotrons were located, rather than solely to the climate scenario. Observations could not be considered fully independent, given the hierarchical structure of the data. Specifically, multiple pears were collected from a single tree, and trees were nested within ecotrons.

Based on the above, a linear mixed model was applied, with climate scenario, pear variety, and location included as fixed effects, and tree and ecotron as random effects. Model assumptions were checked using diagnostic plots. For the linear mixed model, we examined a residuals versus fitted values plot to detect any systematic patterns or heteroscedasticity, and a normal Q-Q plot to assess the normality of the residuals.

As an additional analysis, we also created a binary quality variable. Pears with a quality index of 70 or higher were classified as good quality, while pears below 70 were classified as not good quality. This outcome was analyzed using a logistic mixed model.

2.2.1 Hypotheses for the Quality Index

H_0 : Mean pear quality does not differ across the four climate scenarios,
after accounting for pear varieties, location, and the hierarchical structure of the data.

H_1 : Mean pear quality differs for at least one climate scenario,
after accounting for pear varieties, location, and the hierarchical structure of the data.

2.2.2 Hypotheses for Good Quality Pears

H_0 : The probability of a pear being classified as good quality is the same across the four climate scenarios, after accounting for varieties, location, and clustering.

H_1 : The probability of a pear being classified as good quality differs for at least one climate scenario, after accounting for varieties, location, and clustering.

2.2.3 Linear Mixed Model for Quality Index

$$Y_{ijk} = \beta_0 + \beta_1 \text{Climate2}_{ijk} + \beta_2 \text{Climate3}_{ijk} + \beta_3 \text{Climate4}_{ijk} + \beta_4 \text{Doyenne}_{ijk} + \beta_5 \text{France}_{ijk} + b_{0j} + b_{0k(j)} + \varepsilon_{ijk}$$

where:

- Y_{ijk} : quality index of pear i , from tree k , within ecotron j
- β_0 : expected pear size for Conference pears in Climate Scenario 1 in Belgium for Conference varieties
- $\beta_{1,2,3}$: differences in mean quality for Scenarios 2, 3, and 4 compared with Scenario 1
- β_4 : difference in mean size between Doyenne and Conference pears
- β_5 : difference in mean size between Belgium and France pears
- b_{0j} : random effect for ecotron
- $b_{0k(j)}$: random effect for tree nested within ecotron
- ε_{ijk} : residual error

2.2.4 Logistic Mixed Model for Good Quality

$$\begin{aligned} \text{logit}(P(Y_{ijk} = 1)) = & \beta_0 + \beta_1 \text{Climate2}_{ijk} + \beta_2 \text{Climate3}_{ijk} \\ & + \beta_3 \text{Climate4}_{ijk} + \beta_4 \text{Doyenne}_{ijk} \\ & + \beta_5 \text{France}_{ijk} + b_{0j} + b_{0k(j)} \end{aligned}$$

where:

$$Y_{ijk} = \begin{cases} 1, & \text{if the pear had a quality index of 70 or higher} \\ 0, & \text{otherwise} \end{cases}$$

2.3 Pear Growth Over Time

2.3.1 Study Design and Data Sources

The **Size data** had repeated pear size measurements collected from week 5 to week 24, together with climate scenario, variety, pear ID, tree ID, and ecotron ID. Missing values were removed using `na.omit()` to ensure consistent model fitting and the presence of large sample of pears per climate scenario allowed for that.

2.3.2 Data Structure and Confounding Considerations

The experiment followed a three-level hierarchical structure: Ecotron $\rightarrow$ Tree $\rightarrow$ Pear. In addition, the study included repeated measurements through weekly pear size observations. The following potential confounders and design-related covariates were considered: a) **varieties**: biologically meaningful differences in pear size were expected between Conference and Doyenne pears; therefore, variety was included as an important covariate in the model; b) **location**: all ecotrons were located in Belgium, and therefore no location variability existed.

Statistical Hypotheses

The table 1 shows the hypotheses used to study sub-primary question aimed to evaluate whether different climate scenarios affect pear size over time while accounting for repeated measurements and hierarchical clustering of pears.

Two-sided hypotheses were employed because climate interventions could plausibly either improve or reduce pear growth under future environmental conditions.

Effect	Null Hypothesis (H_0)	Alternative Hypothesis (H_A)
Climate Main Effect	$\beta_{\text{climate}} = 0$ (no baseline climate effect)	$\beta_{\text{climate}} \neq 0$
Climate $\times$ Time	$\beta_{\text{climate} \times \text{week}} = 0$ (climate does not affect growth rate)	$\beta_{\text{climate} \times \text{week}} \neq 0$
Varieties Effect (species)	$\beta_{\text{species}} = 0$	$\beta_{\text{species}} \neq 0$
Species $\times$ Time	$\beta_{\text{species} \times \text{week}} = 0$ (species does not affect growth slope)	$\beta_{\text{species} \times \text{week}} \neq 0$
Species $\times$ Time ²	$\beta_{\text{species} \times \text{week}^2} = 0$ (species does not affect growth curvature)	$\beta_{\text{species} \times \text{week}^2} \neq 0$

Table 1: Statistical hypotheses for the longitudinal pear growth analysis

2.3.3 Model Development

The model equation is:

$$\begin{aligned}
 y_{ij} = & \beta_0 + \beta_1 \text{Climate2}_{ij} + \beta_2 \text{Climate3}_{ij} + \beta_3 \text{Climate4}_{ij} \\
 & + \beta_4 \text{weekc}_{ij} + \beta_5 \text{weekc}_{ij}^2 + \beta_6 \text{Doyenne}_{ij} \\
 & + \beta_7 (\text{Climate2}_{ij} \times \text{weekc}_{ij}) \\
 & + \beta_8 (\text{Climate3}_{ij} \times \text{weekc}_{ij}) \\
 & + \beta_9 (\text{Climate4}_{ij} \times \text{weekc}_{ij}) \\
 & + \beta_{10} (\text{Doyenne}_{ij} \times \text{weekc}_{ij}) \\
 & + \beta_{11} (\text{Doyenne}_{ij} \times \text{weekc}_{ij}^2) \\
 & + b_{0i} + b_{1i} \text{weekc}_{ij} + \varepsilon_{ij}
 \end{aligned}$$

where:

- y_{ij} : pear size for observation j from pear i
- β_0 : expected pear size for Conference pears in Climate Scenario 1 at $\text{week}_c = 0$
- $\beta_{1,2,3}$: difference in mean size between Scenario 2, Scenario 3, and Scenario 4 respectively as compared to Scenario 1 at $\text{week}_c = 0$
- β_4 : linear growth rate over time for Conference pears in Scenario 1
- β_5 : curvature in growth over time for Conference pears
- β_6 : difference in mean size between Doyenne and Conference pears at $\text{week}_c = 0$
- $\beta_{7,8,9}$: linear growth rate in Scenario 2, Scenario 3, and Scenario 4 respectively as compared with Scenario 1
- β_{10} : difference in linear growth rate between Doyenne and Conference pears
- β_{11} : difference in growth curvature between Doyenne and Conference pears
- b_{0i} : random intercept for pear i
- b_{1i} : random slope for centered week for pear i
- ε_{ij} : residual error

Random effects:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{b0}^2 & \sigma_{b01} \\ \sigma_{b01} & \sigma_{b1}^2 \end{pmatrix} \right]$$

and:

$$\varepsilon_{ij} \sim N(0, \sigma^2)$$

Several candidate mixed-effects models were explored, including:

- linear time models with random intercepts only,
- linear time models with random intercepts and random slopes,
- quadratic time models with random intercepts only,
- quadratic time models with random intercepts and random slopes,
- reduced quadratic interaction models,
- and centered-time quadratic models with selective interaction structures.

The selected final model included centered week terms because both `week` and `week`² were included in the model. These terms were highly correlated, potentially leading to multicollinearity. Such multicollinearity destabilized coefficient estimates, and prevented model convergence.

To address this issue, the week variable was centered around its mean:

$$\text{week_c} = \text{week_num} - \overline{\text{week_num}}$$

To assess numerical stability and convergence robustness, the final model was refitted using the `bobyqa` optimizer. For model selection, both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were used. The AIC evaluates model fit: $AIC = -2\log(L) + 2k$, where L represents the likelihood and k represents the number of estimated parameters. Lower AIC values indicate a better balance between goodness-of-fit and model parsimony.

The BIC penalizes model complexity more strongly: $BIC = -2\log(L) + k\log(n)$, where n represents the sample size. Lower BIC values indicate a better model fit while favoring simpler models more strongly than the AIC, particularly in large datasets.

Moreover, the intraclass correlation coefficient (ICC) was calculated to quantify the proportion of total variability in pear size attributable to between-pear differences, thereby assessing the degree of within-pear correlation in the repeated longitudinal measurements.

2.4 Climate-soil-production relations

The analysis is framed as a mechanistic exploratory analysis. The assumed causal structure is presented in fig. 1

Under this interpretation, the climate scenario may affect pear outcomes directly, and may also affect them indirectly through changes in soil properties. A possible causal pathway from climate to trees, and from trees to soil is not considered in this analysis, but it could be in future studies.

We first computed Spearman correlations between final soil variables and pear outcomes separately within each climate scenario. These scenario-specific correlations are useful as

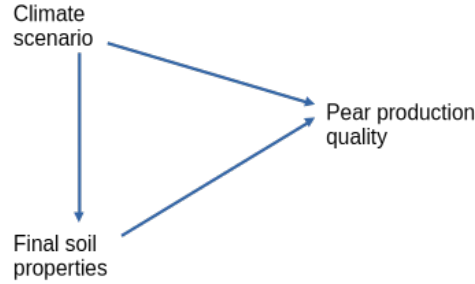


Figure 1: Assumed causal structure for climate scenarios, soil conditions and pear production.

descriptive analyses of the association between final soil state and pear outcomes under each climate condition.

We performed a PCA on the pooled (all ecotrons) final soil variables. To further summarize final soil profiles, k-means clustering was applied to the first two principal components, pooling all soil observations across climate scenarios. The objective of this analysis is to identify groups of similar final soil states and to examine how these soil states are distributed across climate scenarios. Boxplots were produced for outcomes by soil cluster. Kruskal-Wallis tests were performed to assess differences in outcomes between clusters within scenarios.

Because the soil-outcome association may differ by climate scenario, another PCA of the final soil variables was also performed separately within each climate scenario and graphically combined with the values of the outcomes and the ecotron ID. Besides some soil outliers and the fact that final soil conditions can be diverse within one ecotron, no obvious visual pattern was seen for the distribution of quality outcomes along the first two components so this result is not reported here.

3 Result

3.1 Exploratory Data Analysis (EDA)

The 144 trees in the study produced between 5 and 45 pears each (overall mean 19.7, SD 7.8). Figure 3 shows the distribution of pear yield per tree broken down by climate scenario.

Scenario 2 shows a visibly higher median yield and a wider spread compared to the other scenarios. Scenarios 1 and 4 appear nearly identical in their distributions, while Scenario 3 falls in between.

The raw variance-to-mean ratio across all 144 trees is approximately 3.06, which at first sight suggests over-dispersion. However, this marginal calculation ignores the strong between-group heterogeneity induced by climate: trees assigned to different scenarios are

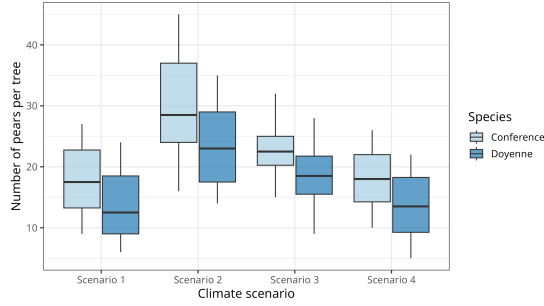


Figure 2: Pear yield per tree by climate scenario and variety.

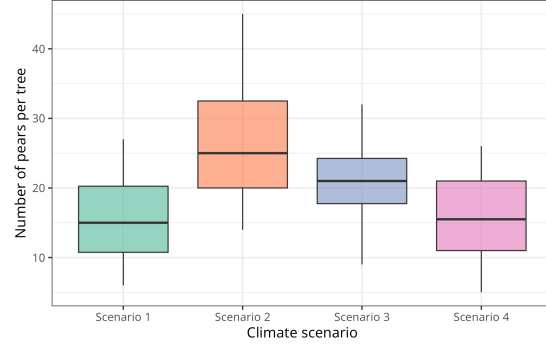


Figure 3: Pear yield per tree by climate scenario. Boxes show the interquartile range; horizontal lines indicate the median; whiskers extend to $1.5 \times \text{IQR}$.

expected to differ systematically in yield, inflating the marginal variance. The within-group dispersion, assessed after model fitting, shows no evidence of extra-Poisson variation (Section 3.2).

Figure 2 reveals that *Conference* pears consistently outyield *Doyenne du Comice* across all four scenarios, and that the ranking of scenarios ($S2 > S3 > S1 \approx S4$) is similar for both varieties, suggesting no strong varieties–climate interaction.

3.2 Model Fit and Assumption Checking

The Poisson GLMM described in the Methods was fitted with `glmer()`. The model produced a boundary (singular) fit warning, indicating that the estimated ecotron random-effect variance equals zero in the full model. This is an expected and interpretable result: because climate is assigned at the ecotron level, once climate is included as a fixed effect it absorbs all systematic between-ecotron variation, leaving no residual ecotron variance to estimate. Crucially, the random intercept remains essential for constructing the likelihood ratio test (see below): without it, the null model would be misspecified and the test statistic inflated.

The Pearson goodness-of-fit ratio for the fitted model is $\hat{\chi}^2/\text{df} = 1.021$ (deviance ratio = 1.045), both very close to 1. There is therefore *no evidence of residual over-dispersion* in the Poisson model after conditioning on the fixed and random effects. As a robustness check, a Negative Binomial GLMM was fitted; its estimated dispersion parameter $\hat{\theta} \approx 185,000$ confirms that the variance is indistinguishable from the mean, so the Negative Binomial model collapses to Poisson ($\text{AIC}_{\text{Poisson}} = 840.6$ vs. $\text{AIC}_{\text{NB}} = 842.6$).

Residual diagnostics (Pearson residuals versus fitted values and a normal Q–Q plot) showed no systematic patterns, heteroscedasticity, or substantial departures from normality, supporting the adequacy of the Poisson GLMM.

3.3 Effect of Climate Scenario on Pear Yield

Overall test

The likelihood ratio test comparing the full model to the model without climate fixed effects gives $\chi^2(3) = 47.02$, $p < 0.001$, providing strong evidence that climate scenario significantly affects pear yield (H_0 in (3) is rejected).

Estimated marginal means

Table 2 reports the estimated marginal mean pear count per tree for each scenario, averaged over varieties and location (back-transformed from the log scale via `emmeans`).

Table 2: Estimated marginal mean pear yield per tree by climate scenario (Poisson GLMM; 95% Wald confidence intervals, back-transformed).

Scenario	Mean pears/tree	95% CI lower	95% CI upper
Scenario 1 (reference)	15.2	14.0	16.5
Scenario 2	26.0	24.4	27.7
Scenario 3	20.0	18.6	21.5
Scenario 4	15.5	14.3	16.9

Figure 4 provides a visual summary of these estimates.

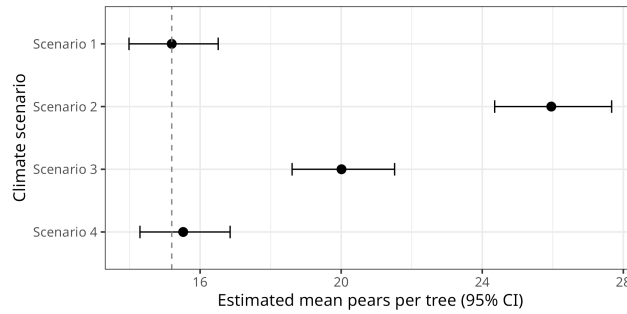


Figure 4: Estimated marginal mean pear yield per tree with 95% confidence intervals for each climate scenario. The dashed vertical line marks the Scenario 1 (reference) estimate.

Pairwise comparisons

Table 3 reports all six pairwise contrasts expressed as rate ratios (Tukey-adjusted).

Scenario 2 (CO₂ removal) yields on average 71% more pears per tree than the no-intervention baseline (Scenario 1; rate ratio = 1.71, 95% CI [1.54, 1.90], $p < 0.001$). Scenario 3 (sustainable energy) also significantly outperforms Scenario 1 by 32% (rate ratio = 1.32, 95% CI [1.18, 1.47], $p < 0.001$). Scenario 4 (transportation) is not distinguishable

Table 3: Pairwise comparisons between climate scenarios. Rate ratios are on the count scale; 95% confidence intervals and p -values are Tukey-adjusted. Ratios > 1 indicate the numerator scenario has higher yield.

Contrast	Rate ratio	95% CI lower	95% CI upper	p -value
S2 / S1	1.71	1.54	1.90	< 0.001
S3 / S1	1.32	1.18	1.47	< 0.001
S4 / S1	1.02	0.91	1.14	0.985
S2 / S3	1.30	1.14	1.47	< 0.001
S2 / S4	1.67	1.46	1.92	< 0.001
S3 / S4	1.29	1.12	1.49	< 0.001

from the reference (rate ratio = 1.02, $p = 0.985$). Furthermore, Scenario 2 significantly outperforms Scenario 3 (rate ratio = 1.30, $p < 0.001$), establishing a clear ordering: $S2 > S3 > S1 \approx S4$.

3.4 Pear Quality Analysis

As shown in Table 4, a total of 2836 pears were included in the analysis. The quality index values ranged from 0 to 100, and the overall mean quality index was 56.12. The descriptive results indicated that pear quality varied depending on the climate scenario.

More specifically, Scenario 1 showed the lowest mean quality index, with a mean value of 49.9. In contrast, Scenario 2 had the highest mean quality index, with a mean value of 60.0, followed by Scenario 3, with a mean value of 58.3. Finally, Scenario 4 was at an intermediate level, with a mean value of 52.9.

Table 4: Descriptive statistics of pear quality across climate scenarios

Climate scenario	Number of pears	Mean quality	SD	Number of good quality pears	Proportion of good quality pears
Scenario 1	562	49.9	23.5	121	21.5%
Scenario 2	960	60.0	24.3	353	36.8%
Scenario 3	740	58.3	25.1	251	33.9%
Scenario 4	574	52.9	24.3	150	26.1%

As shown in Figure 8, which presents the distribution of the quality index by climate scenario, Scenarios 2 and 3 tended to have higher quality values compared to the other scenarios. In contrast, Scenario 1 showed the lowest quality values overall. Scenario 4 appeared to occupy an intermediate position, with some values approaching the threshold for good quality, while the median quality index remained below this threshold.

Figure 9 presents the mean quality index for each climate scenario together with the corresponding 95% confidence intervals. Scenario 2 showed the highest mean quality index, at approximately 60, followed closely by Scenario 3. Scenario 1 had the lowest mean quality index, at around 50. Similar to the previous figure, Scenario 4 again occupied an intermediate position. Overall, these results suggest that pear quality was, on average, higher under Climate Scenarios 2 and 3 compared with Scenario 1.

The third figure (Figure 7) illustrates the proportion of pears classified as good quality under each climate scenario. This figure provides a more practical interpretation of the

results by showing the percentage of pears meeting the predefined good quality threshold.

More specifically, Scenario 2 had the highest proportion of good quality pears at 36.8%, followed by Scenario 3 at 33.9%. Scenario 4 again occupied an intermediate position, with 26.1% of pears classified as good quality. Finally, Scenario 1 showed the lowest proportion of good quality pears, at only 21.5%. The assumptions of the linear mixed model were also checked using diagnostic plots. Moreover, as shown in figure 6 and figure 5, the residuals

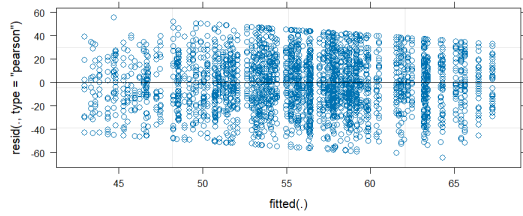


Figure 5: Residual versus fitted values plot for the pear quality mixed-effects model

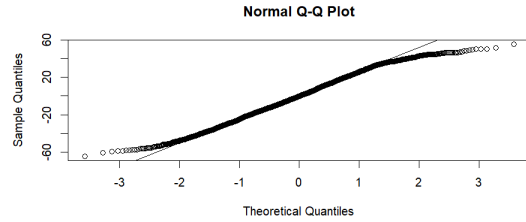


Figure 6: Normal Q-Q plot of residuals for the pear quality mixed-effects model

were roughly centered around zero and did not show a clear systematic pattern. The Q-Q plot showed some deviation in the tails, although this was not unexpected since the quality index is bounded between 0 and 100. Overall, the diagnostic plots did not indicate any serious problem with the use of the linear mixed model.

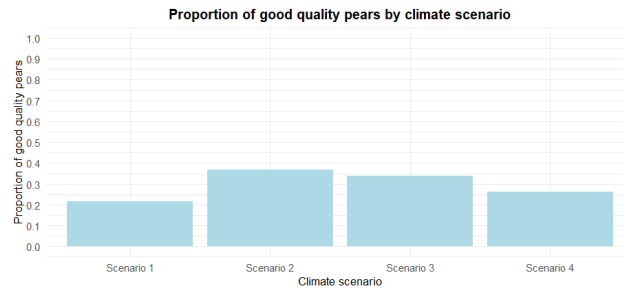


Figure 7: Proportion of pears classified as good quality across climate scenarios

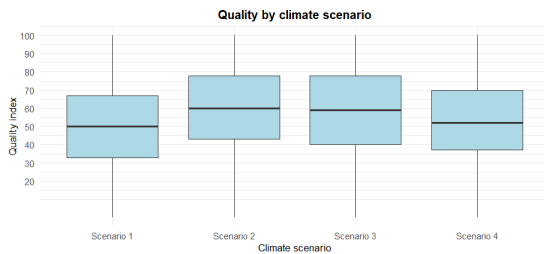


Figure 8: Distribution of pear quality index across climate scenarios

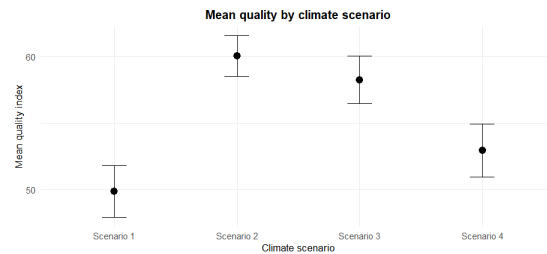


Figure 9: Mean pear quality index with 95% confidence intervals by climate scenario

3.5 Pear Growth with Time

3.5.1 Exploratory Data Analysis (EDA):

Table 5: Average change in mean pear size across climate scenario and varieties

Climate	varieties	Mean Size (Baseline)	Mean Size Difference	SD
Scenario 1	Conference	3.98	5.67	0.41
Scenario 1	Doyenne	3.26	4.66	0.36
Scenario 2	Conference	4.23	6.51	0.53
Scenario 2	Doyenne	3.45	5.66	0.47
Scenario 3	Conference	3.85	5.70	0.44
Scenario 3	Doyenne	3.14	4.63	0.43
Scenario 4	Conference	3.95	5.91	0.39
Scenario 4	Doyenne	3.21	5.21	0.53

Table 5 showed clear differences in pear growth patterns in both climate scenarios and varieties. Conference pears consistently exhibited larger baseline mean sizes and greater average growth over the study period compared to Doyenne pears across all climate conditions. Scenario 2 showed the largest average increase in pear size for both varieties, suggesting that this climate condition may promote faster growth trajectories. In contrast, Scenario 3 generally produced the smallest average growth, particularly for Doyenne pears. Variability in growth, represented by the standard deviation of the differences, remained relatively moderate across groups, indicating fairly consistent growth behavior within each climate–varieties combination.

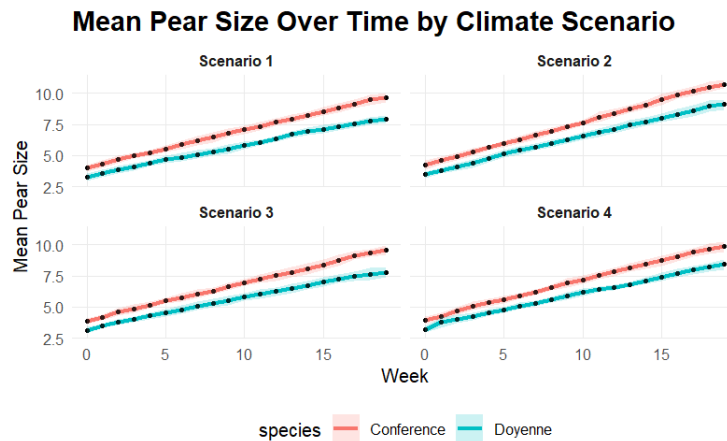


Figure 10: Mean pear size growth over time

Figure 10, showed the average growth pattern of pears across different climate scenarios and fruit varieties. The plots showed approximately linear growth in pear sizes, with Conference pears showing larger baseline size and steeper growth trajectories.

3.5.2 Model Selection and Diagnostics

Several linear mixed-effects models with different fixed- and random-effects structures were compared using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Table 6 summarizes the candidate models and their corresponding fit statistics.

Table 6: Comparison of candidate mixed-effects models using AIC and BIC

Model Structure	Parameters	AIC	BIC
Linear time + random intercept	18	754.7098	865.2879
Linear time + random intercept and slope	20	435.4165	558.2810
Quadratic time + random intercept	26	823.8196	983.5435
Quadratic time + random intercept and slope	28	496.3758	668.3861
Reduced quadratic interaction model	19	414.5225	531.2438
Centered time quadratic model (final model)	16	368.4231	466.7148

The final selected model (Model 6) produced the lowest AIC and BIC values, indicating the best balance between model fit and model complexity among the candidate models. The lower BIC value particularly suggested that the additional terms included in the more complex models were unnecessary.

Model convergence and stability were additionally evaluated using an alternative optimizer (`bobyqa`). Parameter estimates, standard errors, and variance components remained nearly identical to the original fit, confirming numerical stability and robustness of the model estimates. No convergence warnings, singular fits, or optimizer-related issues were detected.

Residual diagnostics indicated that the assumptions of the linear mixed-effects model were reasonably satisfied. The residual-versus-fitted plot showed no major evidence of heteroscedasticity or systematic patterns, suggesting approximately constant residual variance across fitted values. The normal Q–Q plot (Figure 11) demonstrated that residuals followed an approximately normal distribution, with only mild deviations in the tails, especially, at larger fitted values.

The estimated intra-class correlation coefficient (ICC) was approximately 0.47, indicating substantial within-pear dependence and supporting the use of a mixed-effects modeling framework to account for repeated measurements within pears.

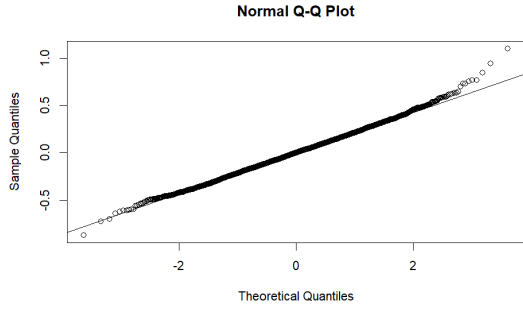


Figure 11: Normal Q–Q plot of growth model residuals

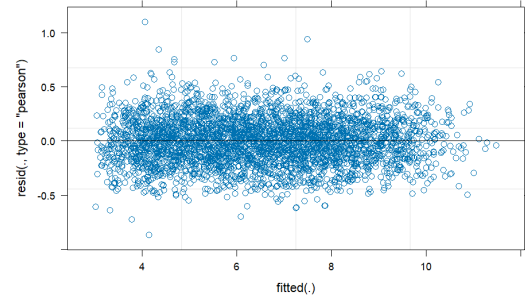


Figure 12: Residual versus fitted values plot for the final growth mixed-effects model

3.5.3 Model Summary

Table 7: Fixed effects estimates from the linear mixed model

Effect	Estimate	Std. Error	df	t value	p-value
Intercept	6.8950	0.0383	175.3000	180.0990	< 0.0001
Climate Scenario 2	0.6507	0.0508	167.0000	12.8100	< 0.0001
Climate Scenario 3	-0.0956	0.0471	167.0000	-2.0280	0.0441
Climate Scenario 4	0.1845	0.0467	167.0000	3.9540	0.0001
week	0.2988	0.0031	167.0000	96.1700	< 0.0001
Variety Doyenne	-1.1120	0.0349	189.8000	-31.8320	< 0.0001
$week^2$	-0.0006	0.0002	3094.0000	-3.4600	0.0005
Climate Scenario 2 $\times$ week	0.0506	0.0042	167.0000	12.1360	< 0.0001
Climate Scenario 3 $\times$ week	-0.0006	0.0039	167.0000	-0.1440	0.8854
Climate Scenario 4 $\times$ week	0.0152	0.0038	167.0000	3.9640	0.0001
week $\times$ Variety Doyenne	-0.0507	0.0028	167.0000	-18.2690	< 0.0001
Variety Doyenne $\times week^2$	-0.0007	0.0003	3094.0000	-2.6580	0.0079

Table 8: Random effects estimates from the linear mixed model

Group	Name	Variance	Std. Dev.	Correlation
pear	Intercept	0.0460	0.2145	
pear	week	0.0003	0.0158	0.70
Residual		0.0515	0.2270	

Table 7 presents the fixed-effect estimates from the linear mixed model evaluating pear growth over time across climate scenarios and varieties. Pear size increased significantly over time ($\beta = 0.2988$, $p < 0.0001$), indicating strong positive growth trajectories throughout the study period. Conference pears showed significantly larger average sizes compared to Doyenne pears ($\beta = -1.1120$, $p < 0.0001$ for Doyenne relative to Conference).

Climate scenario significantly influenced pear growth. Compared to Scenario 1, pears in Scenario 2 exhibited substantially larger sizes and faster growth rates over time ($\beta = 0.6507$, $p < 0.0001$; interaction $\beta = 0.0506$, $p < 0.0001$). Scenario 4 also showed significantly increased growth trajectories relative to Scenario 1. In contrast, Scenario 3 produced only a small negative baseline effect and no significant difference in growth rate over time ($p = 0.8854$ for the interaction term).

The quadratic time effect was significant ($\beta = -0.0006$, $p = 0.0005$), suggesting slight curvature in the growth trajectories rather than perfectly linear growth. Additionally, Doyenne pears showed slower growth over time compared to Conference pears, as indicated by the significant negative interaction between varieties and week ($\beta = -0.0507$, $p < 0.0001$).

Random-effects estimates (Table 8) indicated substantial between-pear variability in baseline size ($\sigma^2 = 0.0460$) and smaller variability in individual growth rates over time. The positive correlation ($r = 0.70$) between random intercepts and slopes suggested that pears with larger baseline sizes tended to exhibit faster growth trajectories.

3.6 Multivariate analysis of climate/soil/production relations

In tab.9, the significant Spearman correlations between final soil parameters and production outcomes, split by climate scenario, are shown. It can be seen that several soil parameters show significant correlations with pear outcomes but the the correlations are not found consistently for all scenario and outcome combinations. In the case of infiltration, the sign of the correlation can be positive or negative depending on the climate scenario.

A PCA on the pooled (all ecotrons) final soil variables was performed. With 17 soil parameters, 2 components explains 36% of the variance, 4 components are needed to explain approximately 60% of the variance, and up to 7 components are needed to approach 75%. In Figure 13, the first two principal components are shown. The figure also explores the distribution of final soil characteristics across climate scenarios and varieties. PC1 and PC2 can be interpreted broadly as separating biological and physical soil properties. Microbial

Soil characteristic	under climate	corr. to	spearman r	p-value	signif.
Bulk Density	Scenario 3	mean quality	-0.281	0.0964	.
Bulk Density	Scenario 3	median quality	-0.395	0.0171	*
Earthworms	Scenario 4	mean quality	-0.414	0.0121	*
Earthworms	Scenario 4	median quality	-0.334	0.0466	*
Infiltration	Scenario 3	mean quality	0.283	0.0946	.
Infiltration	Scenario 3	median quality	0.308	0.0676	.
Infiltration	Scenario 4	mean quality	-0.279	0.0992	.
Microbial Biomass C	Scenario 1	pears n	-0.304	0.0719	.
Mineralizable N	Scenario 2	mean quality	0.294	0.0813	.
Particulate Org. Matter	Scenario 4	mean quality	0.345	0.0395	*
Particulate Org. Matter	Scenario 4	median quality	0.280	0.0979	.
Soil Enzymes	Scenario 1	mean quality	0.381	0.0220	*
Soil Enzymes	Scenario 3	mean quality	0.409	0.0132	*
Soil Enzymes	Scenario 3	median quality	0.485	0.0027	**
Soil Enzymes	Scenario 4	pears n	0.375	0.0240	*
Soil Nitrate	Scenario 4	mean quality	0.368	0.0273	*
Soil Nitrate	Scenario 4	median quality	0.311	0.0651	.
Soil pH	Scenario 4	median quality	0.288	0.0887	.

Table 9: Spearman correlations between final soil parameters and production outcomes, split by climate scenario. Only significant correlations with $\text{abs}(r) \geq 0.2$ are shown. Significance: ., *, **: 0.1 0.05 0.01.

biomass C and N, soil respiration, and total organic carbon point roughly in the same direction and are strongly correlated. Infiltration, water holding capacity, soil porosity and soil depth define another part of the soil characteristics.

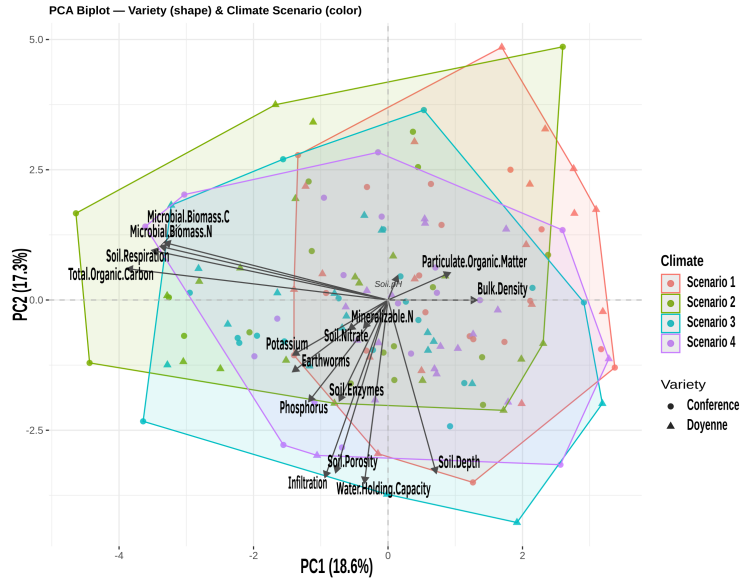


Figure 13: Pooled PCA (all scenarios) for soil parameters: 2 first components. Polygons are added around data points belonging to the same climate scenario. Different shapes are used per variety.

To further summarize final soil characteristics, a k-means clustering applied to the two first PCA soil components (pooling all soil observations across climate scenarios) was performed. The resulting clusters are shown in the two first PCA components in fig.14 along with the associated climate scenario. Fig.15 shows the distribution of those clusters across climate scenarios as a contingency heat map along with a chi-square test p-value.

To explore whether final soil characteristics are associated with pear outcomes boxplots were produced for production outcomes per scenario and soil cluster. In addition, Kruskal-Wallis tests were performed to assess differences in outcomes between clusters within scenarios. However, Kruskal-Wallis tests showed no significant p-value. Fig.16 shows that for scenario 4, which is the one for which one of the clusters has visually different outcomes and which had the lowest p-value (however the apparent difference seems to be driven by only two trees).

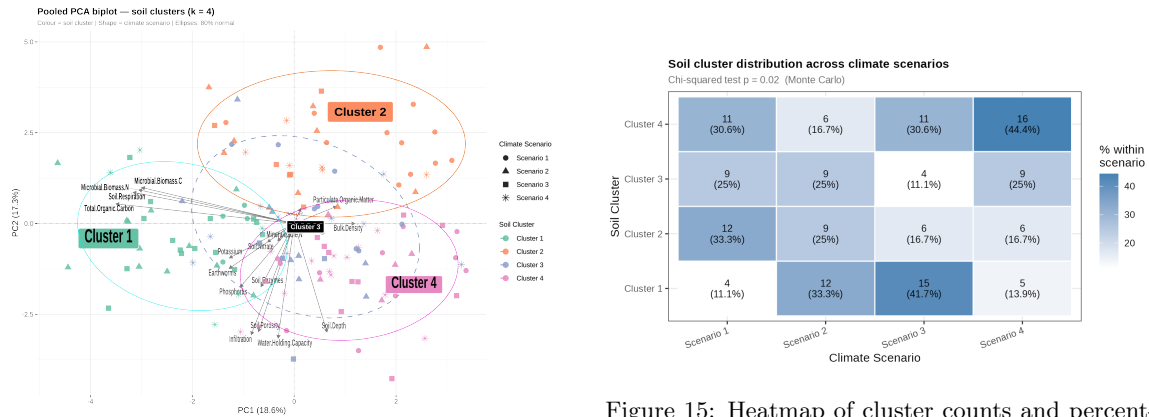


Figure 14: PCA biplot for soil variables. Colour indicates soil cluster, shape indicates climate scenario, and 80% ellipses are drawn around clusters.

Figure 15: Heatmap of cluster counts and percentages per climate scenario. A chi-squared p-value was also computed.

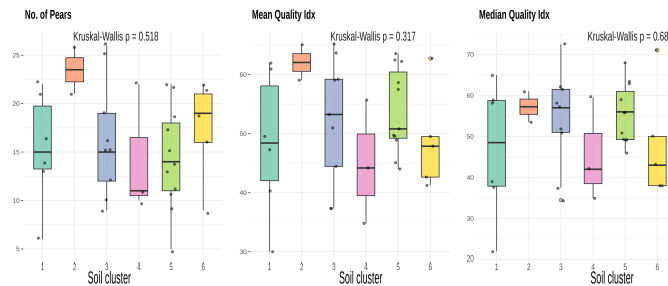


Figure 16: Boxplot comparing production outcomes per final soil k-means cluster for scenario 4 ecotrons.

4 Discussion

4.1 Pear Yield

The results provide clear evidence that climate scenario affects pear yield, with CO₂ removal (Scenario 2) producing the highest output—approximately 71% more pears per tree than the no-intervention baseline—followed by the sustainable energy scenario (Scenario 3, +32%). The transportation scenario (Scenario 4) offers no detectable yield benefit, performing comparably to the no-intervention control. This ordering is consistent across both pear varieties and both locations, suggesting a robust climate effect rather than one driven by a particular subgroup. Taken together with the quality analysis, which finds the same ranking of scenarios for the pear quality index, Scenario 2 emerges as the most favourable future climate for pear production on both dimensions of interest. Scenario 4 is the least promising intervention, providing neither yield nor quality gains over a world without major climate action.

A notable feature of the model is that the ecotron random-effect variance reduces to zero once climate is included as a fixed effect. This reflects the clean experimental design: climate is the only systematic source of between-ecotron variation, so the fixed effect fully explains it. While this produces a boundary singular fit warning, it does not invalidate the analysis; the random intercept remains necessary to correctly specify the null model used in the likelihood ratio test.

Several limitations should be acknowledged. First, the study uses only three ecotrons per scenario per country, which limits the power to detect scenario-specific interactions with varieties or location.

4.2 Pear Quality

Based on all the findings we obtained and observed through our analysis, we conclude that Scenarios 2 and 3 appear to be the best intervention options compared to the other two scenarios. However, it should be noted that the implementation of Scenario 2 is proposed as the most ideal intervention in the ecotrons, as it can offer the highest rates of good quality pears. In contrast, the implementation of climate Scenarios 1 and 4 is not recommended, as it was shown that they cannot deliver high rates of quality pears.

4.3 Pear Growth with Time

The final linear mixed-effects model demonstrated that pear growth changed significantly over time and differed between climate scenarios and varieties. The positive linear effect of time indicated continuous growth throughout the study period, while the significant negative quadratic term suggested that growth trajectories were nonlinear, with growth rates gradually slowing over time, which mimics the biology of life.

The inclusion of random intercepts and random slopes was justified, as pears differed not only in their initial sizes but also in their individual rates of growth over time. The relatively large intra-class correlation coefficient further supported the presence of the strong within-pear dependence.

Climate conditions influenced pear growth primarily through differences in linear growth rates. Pears grown under Climate Scenario 2 exhibited significantly faster growth compared to the reference climate condition, while Scenario 4 also demonstrated moderately increased growth trajectories. In contrast, Scenario 3 behaved similarly to the reference climate, showing no meaningful difference in growth rate over time.

Clear variety differences were also identified. Doyenne pears were significantly smaller on average than Conference pears and exhibited slower growth rates throughout the study period. In addition, Doyenne pears showed stronger growth deceleration, indicating that their growth trajectories flattened more rapidly over time compared to Conference pears.

Model diagnostics indicated that the assumptions of the linear mixed-effects model were reasonably satisfied. Residual analyses suggested acceptable homoscedasticity with only mild deviations at larger fitted values, while the normal Q–Q plot indicated approximately normal residual behavior. Besides, no major convergence or singularity issues were detected.

One limitation of this study is the relatively small number of ecotrons and trees, which limited the ability to reliably estimate higher-level variance components. Although the final model primarily captured variability at the pear level, some unexplained variability may still originate from tree- and ecotron-level environmental differences. This highlights the importance of conducting future studies with larger hierarchical sample sizes to more robustly evaluate higher-level sources of variation.

4.4 Soil as a mediator between climate and pear quality

The GLMM models elaborated here estimates the overall association between climate scenario and pear outcomes, including both direct climate effects and indirect effects that may operate through changes in soil. Soil variables are not to be included in such a model to be able to answer the main research question: the multivariate analysis presented here is rather useful to support two secondary questions: 1. Did the climate scenarios produce different final soil states? 2. Are those final soil states associated with pear production or quality?

Spearman correlations (tab.9) suggest that climate may not only shift the final soil profile, but also change which soil properties are most strongly related to production or quality. The fact that in the pooled PCA only around 37% of the variance is explained by the two first components indicates that final soil conditions are multidimensional. The soil variables do not collapse into one or two dominant variables. Looking at the pooled PCA biplot (fig.13), scenario 1 appears to have a different final soil profile from the other scenarios, this indicates that the climate scenario induced a distinct soil state.

The kmeans clustering distinguished four clusters. The overlapping with the PCA

diagram suggests 3 clusters quite separated (clusters 1,2,4) and a cluster 3 roughly at the middle with a lot of overlaps to the others. It does not show an obvious visual separation of clusters by climate scenario, but the heatmap indicates imbalances in cluster distribution across scenarios. Under our causal interpretation, this indicates that climate scenarios are associated with different final soil states (that cluster imbalance is consistent with the hypothesis of soil a mediator). Clustering helps to describe how the soil system responded to the imposed climates. The boxplot of clusters vs production outcomes, per scenario, and the Kruskal-Wallis tests does not provide however strong evidence for a soil-cluster effect on pear outcomes within scenarios, but it is worth examining it cautiously further.

These analyzes should be interpreted as evidence about possible mechanisms, not as definitive causal estimates of soil effects. Models including final soil variables (e.g. using PCA components, or soil clusters as categorical variables) could be considered as secondary mechanistic analyses (not replacing the primary total-effect model). If the estimated climate effect is reduced after adding final soil variables, this may suggest that part of the climate effect is associated with changes in soil, but this should be interpreted cautiously because mediation analysis requires causal assumptions and sufficient independent replication. In future studies, soil variables should also be measured over time, because soil final state can also be a late consequence (e.g. soil changes may occur too late enough to plausibly influence production outcomes).

References

- [1] Molenberghs, G. and Verbeke, G. (2005). *Models for Discrete Longitudinal Data*. Springer, New York.
- [2] Bates, D., Mächler, M., Bolker, B. and Walker, S. (2015). Fitting linear mixed-effects models using *lme4*. *Journal of Statistical Software*, **67**(1), 1–48.
- [3] Lenth, R.V. (2024). *emmeans*: Estimated marginal means, aka least-squares means. R package version 1.10.0. Available at: <https://CRAN.R-project.org/package=emmeans>.
- [4] Vandendijck, Y. (2025-2026). *Sample Size Calculation in R*. Course notes, Universiteit Hasselt.

Appendix - R code

A R Code for the Main Research Question Analysis

```
# Poisson GLMM
m_pois <- glmer(
  pears_n ~ climate + species + location + (1 | ecotr_id),
```



```

    family = poisson(link = "log"),
  )

# Over-dispersion check
dev_ratio    <- deviance(m_pois) / df.residual(m_pois)
pearson_ratio <- sum(residuals(m_pois, type = "pearson")^2) / df.residual(m_pois)

# Ecotron random effect: full vs. null model
m_pois_null <- glmer(
  pears_n ~ species + location + (1 | ecotr_id),
  family = poisson(link = "log"),
)

# Negative Binomial robustness check
m_nb <- glmer.nb(
  pears_n ~ climate + species + location + (1 | ecotr_id),
  data = tree_counts
)
AIC(m_pois, m_nb)

# Residual diagnostics
pearson_res <- residuals(m_pois, type = "pearson")
fitted_vals <- fitted(m_pois)
abline(h = 0, lty = 2, col = "red")
qqnorm(pearson_res); qqline(pearson_res, col = "red")

# Likelihood Ratio Test for climate
lrt_result <- anova(m_pois_null, m_pois)

# Fixed effects and confidence intervals
fe <- fixef(m_pois)
ci <- confint(m_pois, method = "Wald", parm = "beta_")

# Estimated marginal means and pairwise contrasts
emm <- emmeans(m_pois, ~ climate, type = "response")
pairs(emm, adjust = "tukey") |> summary(infer = TRUE)

```

B Pear Quality mixed models

```

m2 <- lmer(quality_idx ~ climate + species + location +
  (1 | ecotr_id) + (1 | ecotr_id:tree_id),
  data = outcome_clean)

fixef(m2)
confint(m2, method = "Wald")

# Test the climate effect for quality
m2_null <- lmer(quality_idx ~ species + location +

```

```

      (1 | ecotr_id) + (1 | ecotr_id:tree_id),
      data = outcome_clean,
      REML = FALSE)

m2_ml <- update(m2, REML = FALSE)
anova(m2_null, m2_ml)
m3 <- lmer(quality_idx ~ climate + species + location + climate:species +
      (1 | ecotr_id) + (1 | ecotr_id:tree_id),
      data = outcome_clean)
anova(update(m2, REML = FALSE), update(m3, REML = FALSE))

# Good quality = quality index >= 70
m_good <- glmer(good_quality ~ climate + species + location +
      (1 | ecotr_id) + (1 | ecotr_id:tree_id),
      data = outcome_clean,
      family = binomial)

# Test the climate effect for good quality
m_good_null <- glmer(good_quality ~ species + location +
      (1 | ecotr_id) + (1 | ecotr_id:tree_id),
      data = outcome_clean,
      family = binomial)
anova(m_good_null, m_good, test = "Chisq")

```

C Candidate models considered for pear growth analysis

All five candidate models share the general form

$$y_{ij} = \mathbf{x}_{ij}^\top \boldsymbol{\beta} + b_{0i} + b_{1i}^* \text{Week}_{ij} + \varepsilon_{ij},$$

where b_{1i}^* is included only in models with a random slope (Table 10). The fixed-effects vector $\mathbf{x}_{ij}^\top \boldsymbol{\beta}$ contains Climate, Species, the time polynomial, and their interactions as detailed below.

Table 10: Summary of candidate models. “Full interactions” denotes all two- and three-way terms among Climate, Species, and the time covariates; “Reduced” drops the three-way and Climate $\times$ Species terms.

Model	Time polynomial	Interactions	Random effects
1	Week	Full	b_{0i}
2	Week	Full	b_{0i} , b_{1i} Week
3	Week, Week ²	Full	b_{0i}
4	Week, Week ²	Full	b_{0i} , b_{1i} Week
5	Week, Week ²	Reduced	b_{0i} , b_{1i} Week

D Model Convergence Check Using Alternative Optimizer

R code

```
# Refit the final selected model using the bobyqa optimizer
model\_check <- update(
  model6,
  control = lmerControl(optimizer = "bobyqa")
)
```

E R code for the multivariate soil analysis

Spearman correlations

```
if (n < 5) {
  r <- NA; pval <- NA
} else {
  ct <- cor.test(x[ok], y[ok], method = "spearman", exact = FALSE)
  r <- ct$estimate
  pval <- ct$p.value
}
```

k-means clustering and chi-squared test

```
set.seed(42)
sil_vals <- numeric(K_MAX - K_MIN + 1)
for (k in K_MIN:K_MAX) {
  km <- kmeans(pc_scores, centers = k, nstart = 50, iter.max = 200)
  sil <- silhouette(km$cluster, dist(pc_scores))
  sil_vals[k - K_MIN + 1] <- mean(sil[, 3])
}
k_best <- sil_profile$k[which.max(sil_profile$avg_sil)]
km_final <- kmeans(pc_scores, centers = k_best, nstart = 200, iter.max = 300)
chi_result <- chisq.test(ct, simulate.p.value = TRUE, B = 5000)
```

Kruskal–Wallis test on outcomes by soil cluster

```
kw <- kruskal.test(df_cl[[ov]] ~ df_cl$soil_cluster)
```